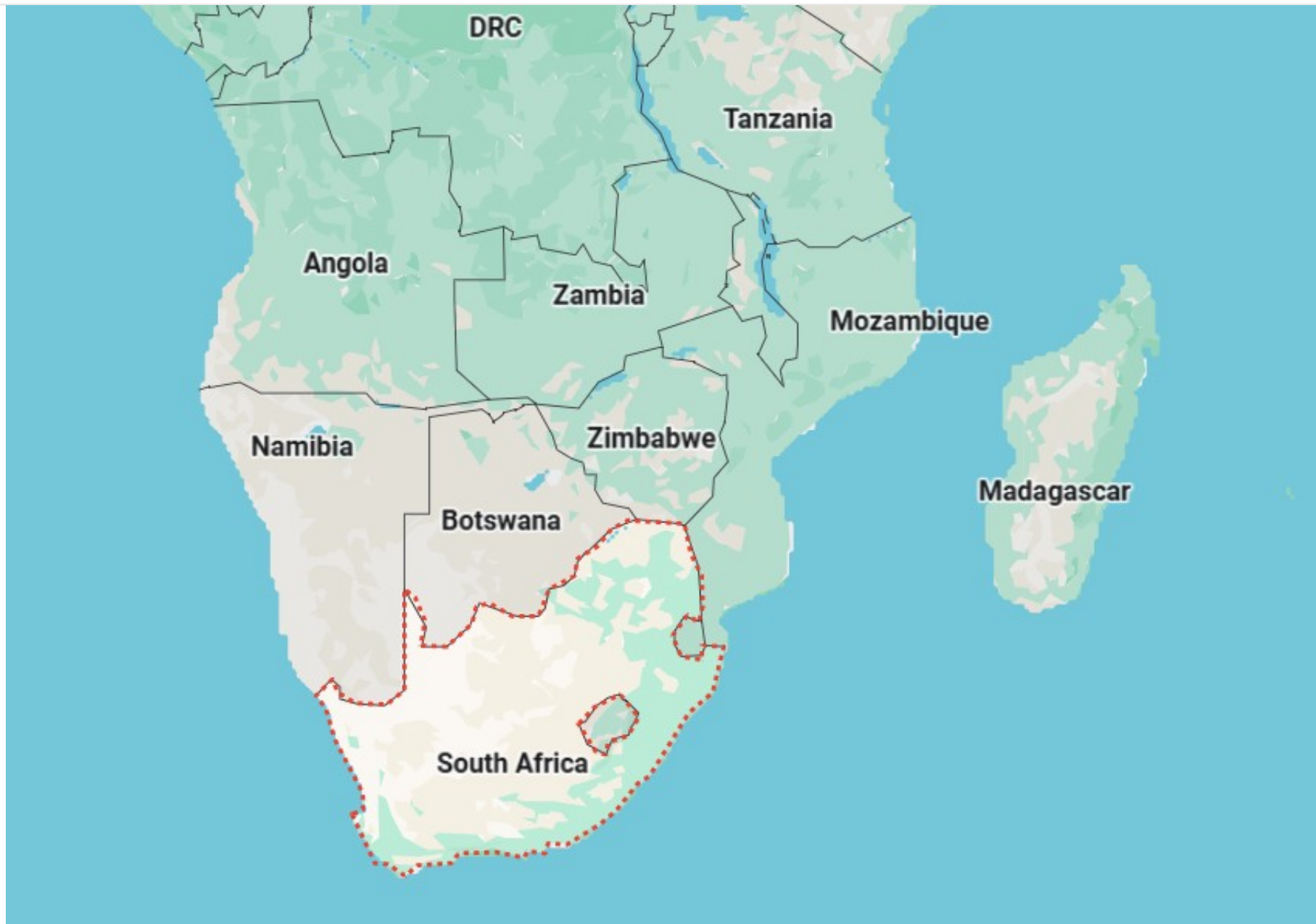


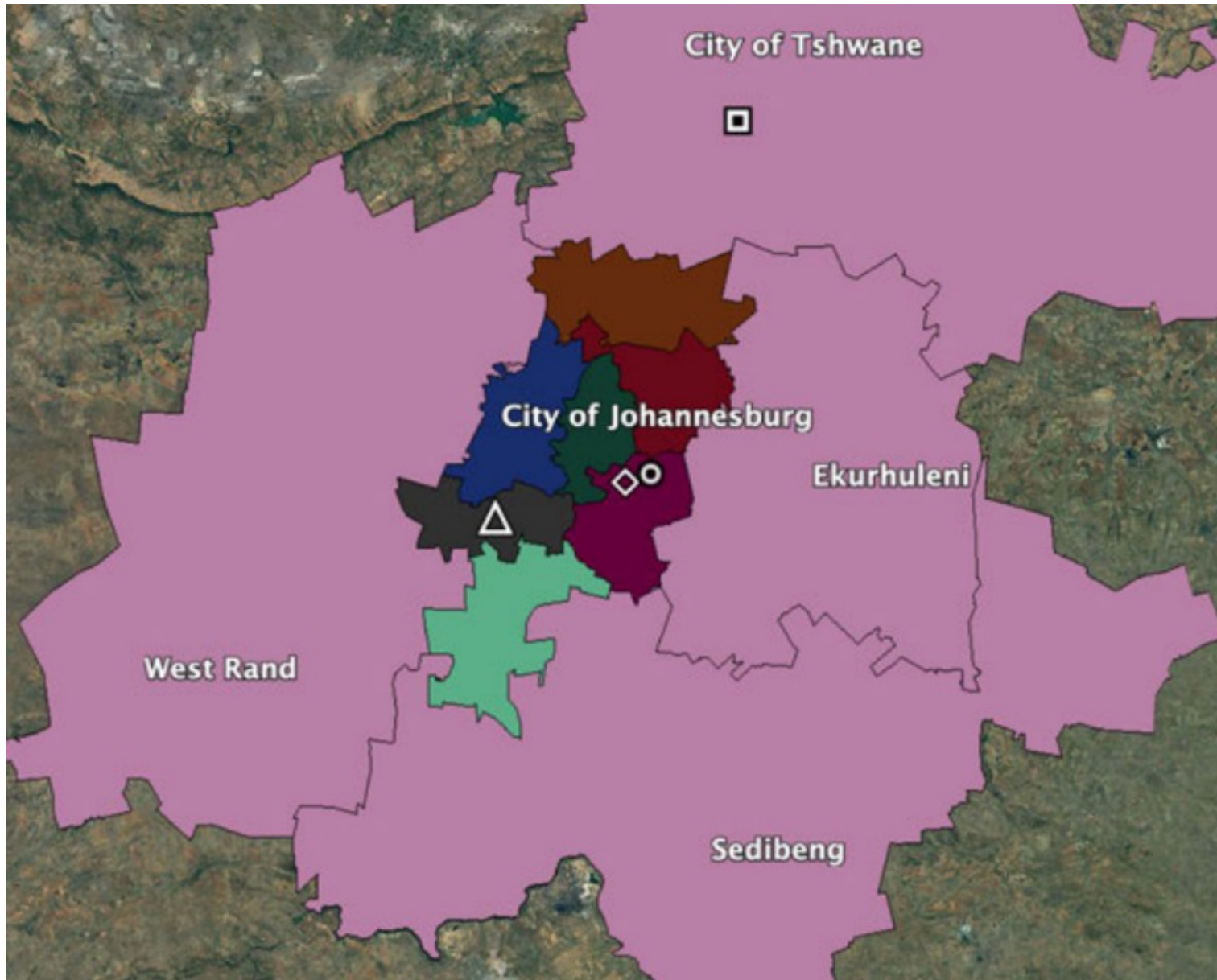


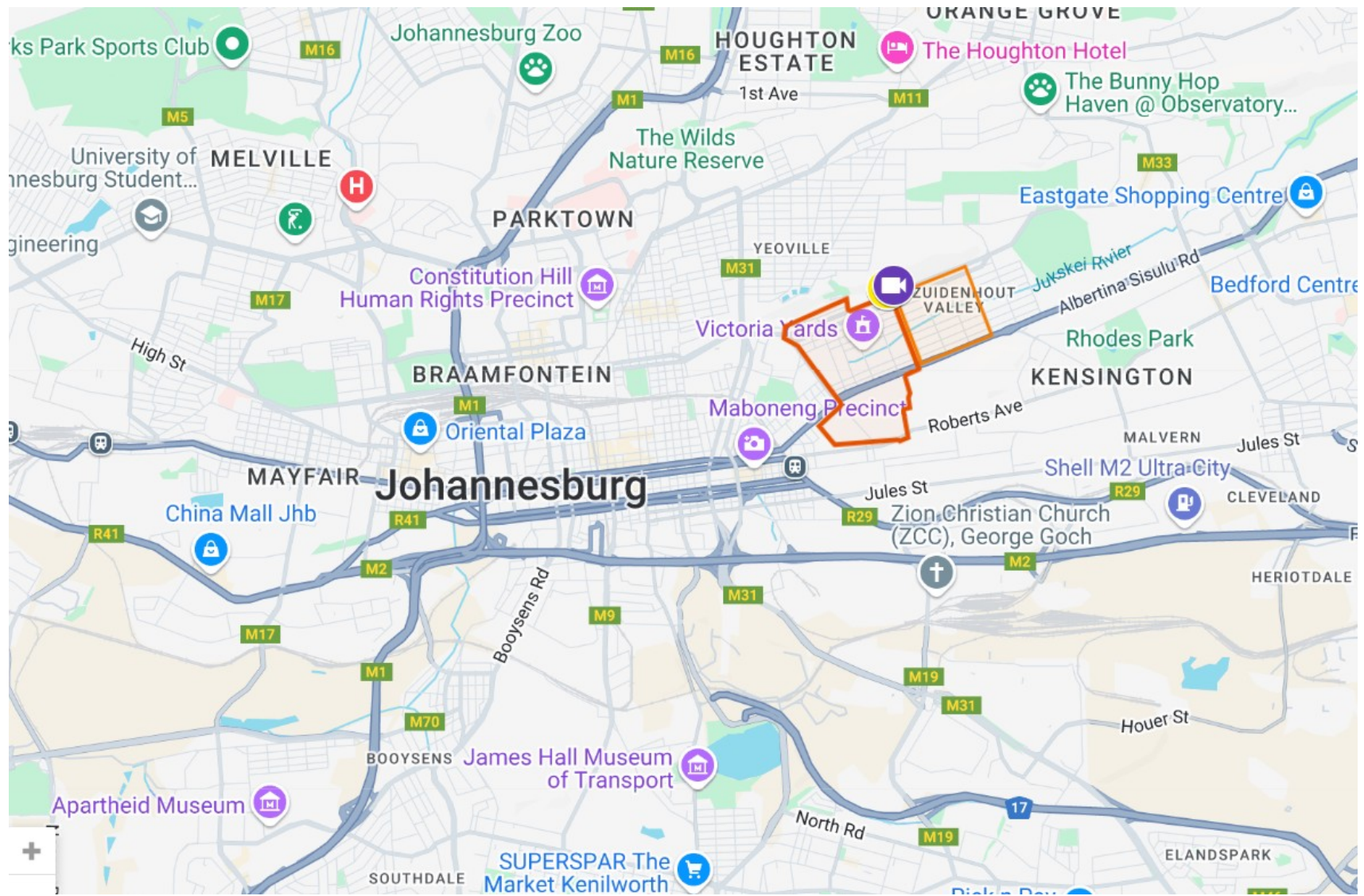
Makers Valley, South Africa

Sally Earl

Materials by EWB









South African History

- South Africa has some of the oldest human fossils in the world
- Some hominin fossils found were over 3.5 millions year old in an area called 'The Cradle of Humankind'
- Became a self-governing state in 1934



Modern-Day South Africa

- Parliamentary republic
- South Africa has an incredibly diverse culture
- Designated as a UN Developing Economy
- One of the most unequal economies in the world, with the highest Gini coefficient



Demographics

- South Africa's Population: 63,015,904
- Johannesburg's Population: 6,300,000
- Makers Valley Population: 46,000

Demographics

- Johannesburg's Population Density is 2900 people per km²
 - Makers Valley is an inner-city region
- In contrast:
 - The UK's population density is 274 people per km²
 - England's population density is 434 people per km²
 - Norwich's population density is 3690 per km²



Makers Valley

- 44% of the population is aged 15-34 years old
- Approximately 60% of residents were born outside of South Africa
- The area is historically disadvantaged, but some regeneration is starting



Makers Valley

- A hub of the maker movement, which focuses on creating things, innovation, and digital design
- Diverse area: 12 languages are official in South Africa



Housing

- There is insufficient housing in the area
- Some houses are home to multiple families
- Many people live in 'hijacked' abandoned buildings, which can have large numbers of people living in them



Infrastructure: Water

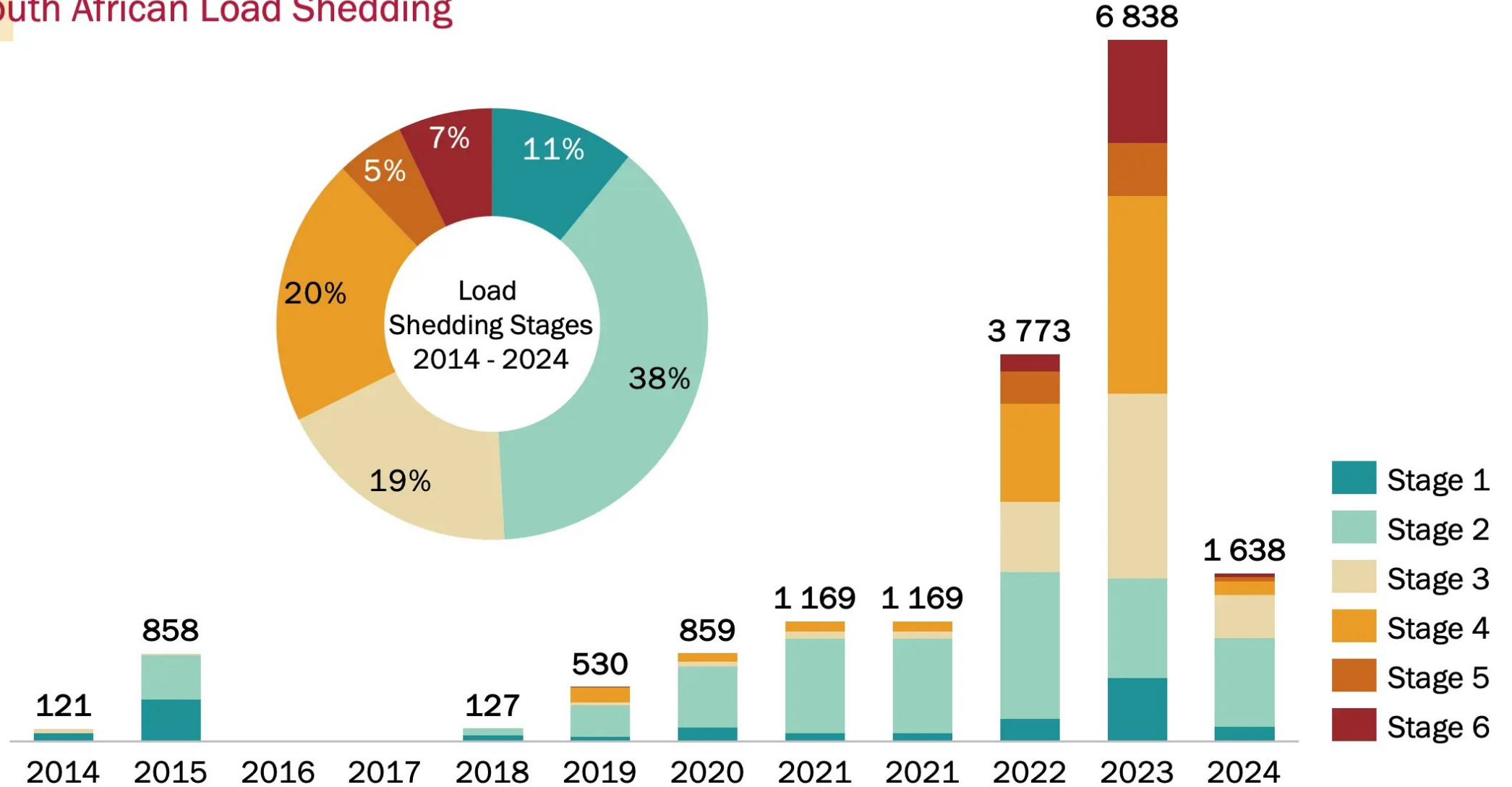
- Water shortages are common, and likely to become an increasing issue
- Around 45% of water supplied is lost to infrastructure leaks and theft
- Water is supplied through pipes and pumps into most households
- Water is expensive!

Infrastructure: Electricity

- Most households have electricity through prepaid meters
- The electricity supply is unreliable
 - 'Load shedding' - the practice of controlled shutdown of the electricity supply is common

South African Load Shedding

Hours of National Load Shedding



The Centre for Renewable and Sustainable Energy Studies (CRSES) | Stellenbosch University

Source: Eskom 2025c; EskomSePush 2024.

<https://www.crses.sun.ac.za/sa-energy-stats/>

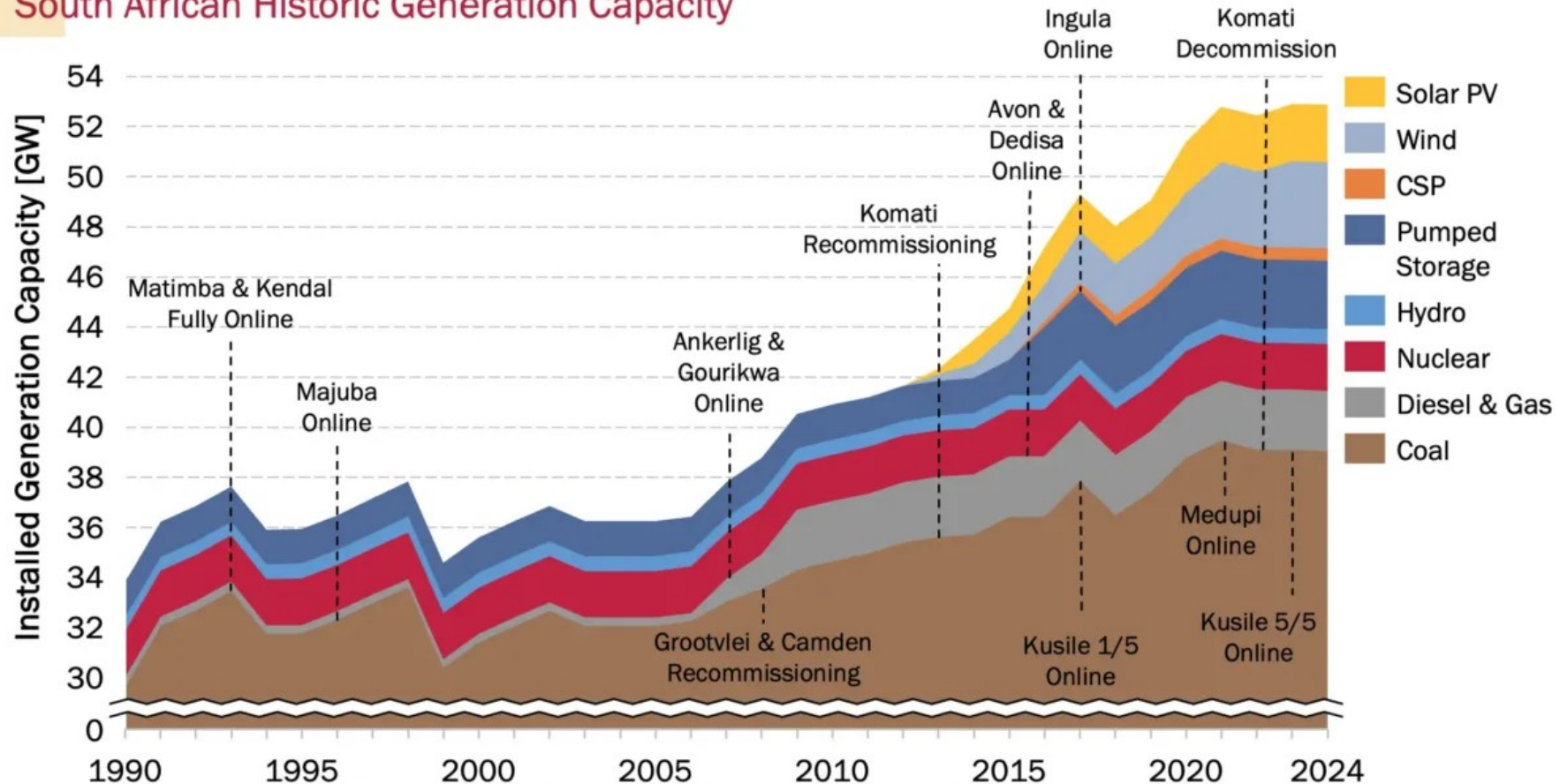


Technology

- Most people have phones with 4G access
- Some residents have 2 phones: due to the risk of theft they leave their smartphone at home
- Computer ownership is less common

The Impact of Climate Change

South African Historic Generation Capacity



The Centre for Renewable and Sustainable Energy Studies (CRSES) | Stellenbosch University

Source: Design as per Fraunhofer ISE 2021 and CSIR 2023; Eskom 2025c. Notes: The status of power stations over 300MW are labelled.



The Impact of Climate Change

- Water shortages are getting worse
- Increased storms means more flooding
- Water shortages also mean urban farming is more difficult

UN Sustainable Development Goals



Areas to Consider

Food

Transport

Digital

Built
Environme
nt

Waste

Energy

Water

Sanitation